

AQUA-SHIELD

Acoustic intelligence for underwater anomaly, debris & marine-hazard localization from side-scan sonar.

- Detect
- Verify
- Localize
- Act

The insight that shapes the whole system: 74% of our sonar frames contain no target at all. Precision — not recall — is the binding constraint.

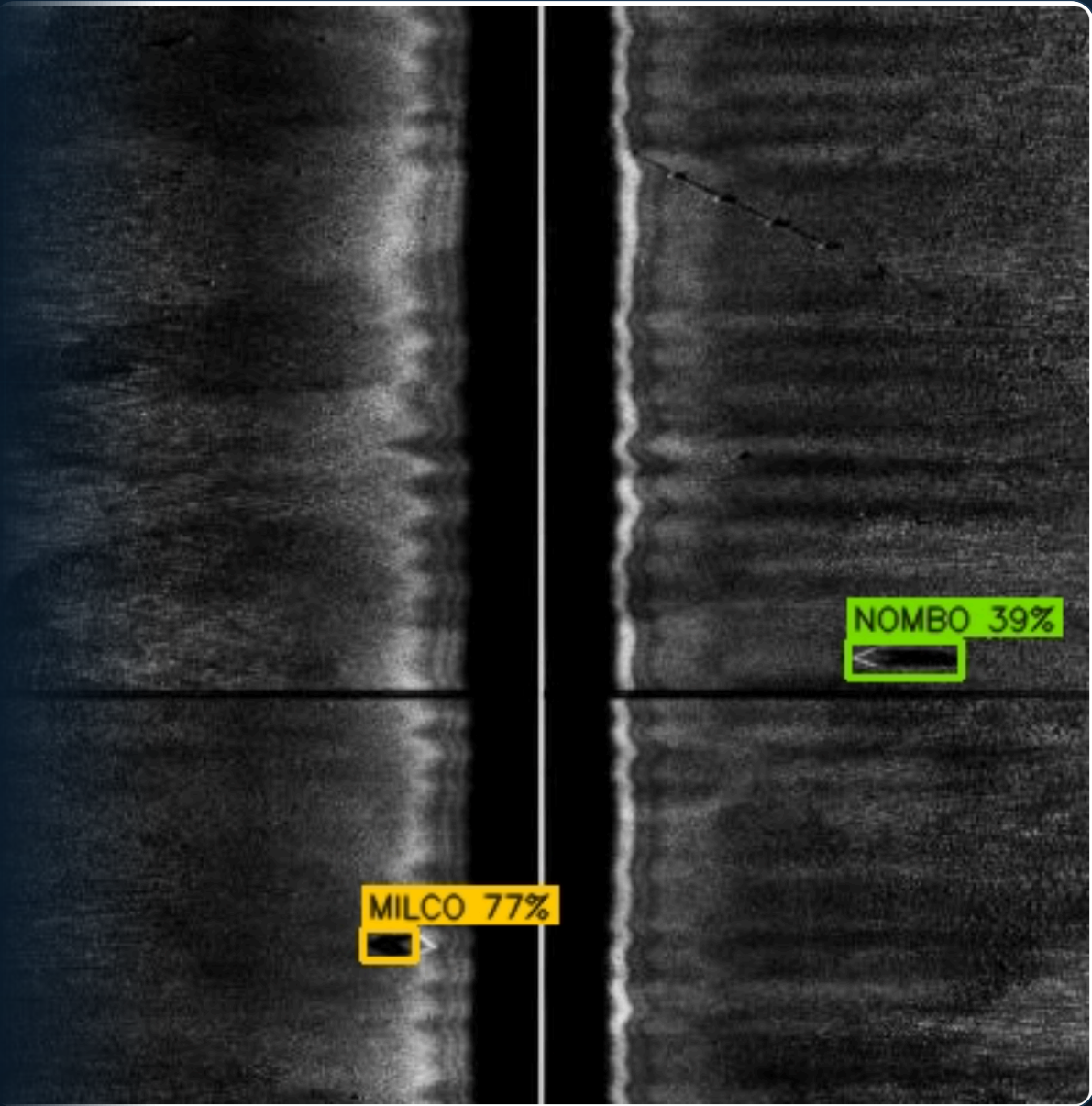
PROBLEM STATEMENT ID
26057

PS CATEGORY
Software

THEME
Disaster Management

TEAM ID · NAME
Your Team ID · Your Team Name

Problem Statement — AI-powered automated underwater marine debris and anomaly detection using side-scan sonar imagery · Ministry of Earth Sciences / NIOT



LIVE SYSTEM OUTPUT

Real held-out survey frame 0460_2018, processed by the running prototype. Yellow = man-made, green = ambiguous — both accepted by the verification stage before display.

Why the obvious approach fails here

Everyone optimises recall. On this data that is the wrong objective — and it is why sonar AI does not reach operators.

THE PROBLEM

- Derelict fishing gear — **ghost nets** — keeps killing after it is lost: entangling marine life, smothering reefs, fouling propellers.
- Finding it means a human reading **thousands of km** of side-scan sonar by eye — slow, fatiguing, inconsistent.
- Seabed clutter — rocks, sand ripples, acoustic shadows — looks exactly like a target.

WHAT EXISTS TODAY — AND ITS LIMIT

SYSTEM	LIMITATION
GhostVision JMSE 2025	Closest prior system. Licence unresolved (NOASSERTION) — cannot be reused.
sidescantools GPL-3.0	Sonar processing only. No detection, no verification.
Generic YOLO detector-only	Fires on empty seabed. No evidence, no calibration, no refusal.

OUR INSIGHT

74%

of frames in our held-out surveys contain **no target at all** (473 of 612 — measured).

A recall-tuned detector alarms constantly on empty seabed, the analyst stops trusting it, and the system is abandoned. **Precision is the binding constraint.**

OUR ANSWER — FOUR THINGS THAT ARE OURS

01

Independent physical verification

10 features measured from the pixels — shadow coherence, contrast, compactness, texture — that do *not* depend on the detector's own opinion. Fitted on a held-out survey.

02

Confidence that means something

Platt calibration fitted on held-out data. When it is not fitted, every hazard is stamped **calibrated: false** — never a silently wrong number.

03

Refusal as a feature

No navigation metadata → no coordinate. **null**, plus the reason. A fabricated latitude exports cleanly to CSV and sends a vessel to open water.

04

Confidence ≠ Priority

“Is it real?” and “should you care?” are different questions, scored separately, so a low-confidence large hazard is not buried.

The verifier is a diagnostic instrument, not just a classifier. Its fitted weights gave acoustic shadow a large *negative* weight — the opposite of the physics. That exposed a real defect: a preprocessing chain applied at inference that the detector had never been trained on. Fixing the mismatch moved F1 from 0.012 → 0.144 (measured) and turned both shadow features positive. A hand-tuned threshold would have hidden it.



The real pipeline — ten stages, every one measured

Not “user → AI → result”. Each stage is separately testable, separately ablated, and can be switched off to measure what it contributes.



DATA — REAL, LICENSED, LEAKAGE-FREE

DATASET	SCALE	SPLIT BY
MILCO / NOMBO CC BY 4.0	465 / 93 / 612 frames · 191 test objects	acquisition year
Derelict crab pot CC BY-SA 4.0	6,674 imgs 9,311 objects · 107 recordings	recording ID

Never a random split. Consecutive sonar frames share seabed, gain settings and often the same object — a random split leaks and inflates every number. Train 2015+2010 → fit 2017 → test 2018+2021, asserted by a test.

CORE ALGORITHM — THE VERIFICATION STAGE

Every detector candidate is re-examined against evidence measured **from the pixels**, independent of the detector score:

shadow ratio

shadow side consistency

local SNR

contrast

compactness

aspect ratio

edge density

relative texture

area fraction

range position

→ **L2-regularised logistic regression**, fitted on the held-out validation survey, with per-detection weight attribution so an operator can see *why* a candidate was accepted or rejected.

→ **Platt calibration** maps the score to a probability; identity transform + explicit flag when unfitted.

SYSTEM & STACK

LAYER	COMPONENT
Interface	Streamlit operator console
Service	FastAPI (REST)
Model	PyTorch 2.13 · MPS / CUDA / CPU
Store	SQLite hazard register
Edge	ONNX Runtime
Geo	pyproj · GeoJSON → QGIS

Backend-swappable detector: the pipeline depends on an interface, not on Ultralytics, so the AGPL-3.0 backend can be replaced without touching the verification, calibration or reporting stages.

I/O CONTRACT

SSS frame 1024×1024 + nav CSV / GeoTIFF

640px tiles

boxes + class + raw score

10-feature vector → p(real)

hazard record

GeoJSON · CSV · SQLite

hazard record = {id, class, level1, confidence, calibrated, priority, priority_band, lat, lon, uncertainty_m, observations, evidence[10], provenance} — every field an operator needs to act, audit, or reject the call.

It runs today — and this is how we will prove it

Left: the actual prototype, actual output, on held-out data. Right: the metrics we commit to hitting, stated as targets, not claims.

HERO DEMO — ONE DETERMINISTIC 90-SECOND RUN

8 held-out frames

›

QC

›

2 raw candidates

›

verify

›

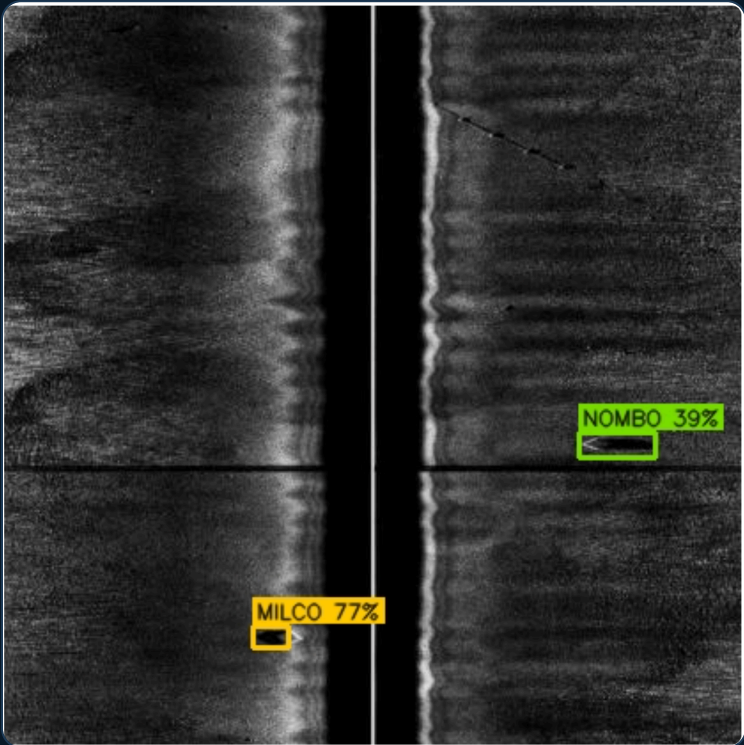
2 hazards

›

1 HIGH

›

export



Frame 0460_2018 — live output of the running system, not a mockup.

WHAT THE OPERATOR IS SHOWN

- MILCO

• mine-like object

confidence 77.0% • band HIGH • priority 58.7
- NOMBO

• bottom object uncertain

confidence 39.5% • band LOW • priority 33.2

Both survived the verification stage. NOMBO is reported as *uncertain* rather than suppressed — the operator decides, with the evidence and a calibrated number in front of them.

hazard_id	class	level1	confidence_%	band	calibrated	priority	priority_band	obs	lat	lon	±m	length_m	frames
AQS-00001	mine_like_object	MAN_MADE	76.99	HIGH	✓	58.7	HIGH	1	None	None	None	None	0460_2
AQS-00002	bottom_object_uncertain	AMBIGUOUS	39.48	LOW	✓	33.2	ROUTINE	1	None	None	None	None	0460_2

Refusal, visible in the product: this survey ships no navigation, so lat / lon / ±m come back None — the system declines to invent a coordinate.

TARGET VALIDATION — WHAT WE WILL PROVE AT THE FINALS

METRIC THAT MATTERS	MEASURED TODAY	TARGET
False-alarm rate on target-free frames	25 / 473 (5.3%)	≤ 2%
Precision @ IoU 0.3	0.322	≥ 0.60
Recall @ IoU 0.3	0.099	≥ 0.45
Ghost-gear detection, mAP50	0.323 raw, 1 run	≥ 0.55
Confidence calibration error (ECE)	fitted, not yet reported	≤ 0.05
Geolocation error (CEP)	not measurable — no nav in available data	≤ 10 m
Inference latency / frame	21.4 ms MPS • 8.5 ms ONNX	≤ 30 ms edge
Sustained throughput	37.4 frames/s	≥ 30 f/s on edge

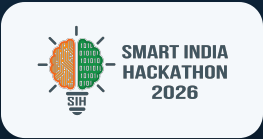
Measured = reproducible from experiments/ on held-out surveys. Target = a goal, not a result. We publish both columns so the gap is visible rather than hidden.

BASELINES WE WILL MEASURE AGAINST

detector-only (no verification) • hand-written rule filter • YOLO11s / RT-DETR at equal compute • published SSS detectors on their own reported splits.

RISKS — AND THE MITIGATION ALREADY BUILT

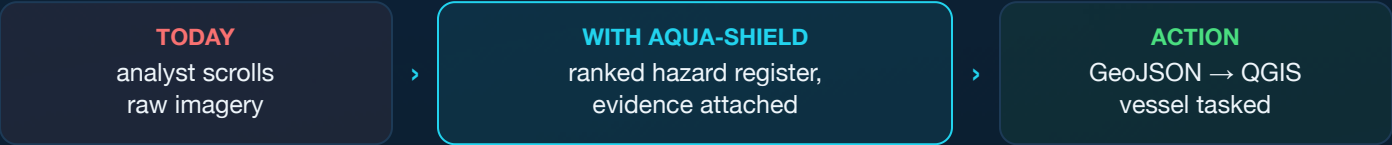
Scarce coastal SSS labels	leakage-free recording-level splits; honest reporting
Detector backend is AGPL-3.0	interface-isolated; ONNX path is backend-free
Speckle noise degrades detection	noise-robust checkpoint trained + shipped separately



From raw sonar to a tasked cleanup vessel

The deliverable is not a model file. It is a ranked, geolocated, auditable hazard register an operator can act on the same day.

WHAT CHANGES OPERATIONALLY



- Every hazard carries its **evidence, calibrated confidence and provenance** — an operator can audit a decision, not just accept it.
- Repeat sightings are merged into unique hazards; positional uncertainty tightens by $\sim\sqrt{N}$ over repeat fixes.
- We do **not** claim a percentage of analyst time saved. That requires a user study we have not run.

hazard_id	lat	lon	class	confidence_pct	priority	priority_band	uncertainty_m	observations
AQS-00001	12.9208	80.3399	mine_like_object	39.08	53.9	ELEVATED	6.01	1
AQS-00002	12.9206	80.3399	mine_like_object	37.06	54.4	ELEVATED	5.99	1
AQS-00003	12.9199	80.3404	mine_like_object	61.69	60.8	HIGH	6	1
AQS-00004	12.9207	80.3406	mine_like_object	37.57	53	ELEVATED	5.93	1

The exported register, from the running system: real coordinates, a per-hazard uncertainty in metres, and a priority band — this is what a cleanup vessel is tasked from.

WHO IT SERVES

BENEFICIARY	BENEFIT
MoES / NIOT survey & cleanup ops	Direct fit to the problem statement's own agency; runs on survey hardware already aboard.
Fisheries & coastal management	Ghost gear keeps killing catch and habitat long after loss; early detection breaks that cycle.
Ports & navigation safety	Mine-like and man-made bottom objects flagged and prioritised before they foul a channel.
Environmental research	Auditable, licence-clean dataset and reproducible experiment registry.

Deployment context. The georeferenced run above places hazards at 12.92°N, 80.34°E — off the Tamil Nadu coast, NIOT's own operating area. The navigation track in that demo is **synthetic**, and the product says so on screen: we are showing the geolocation path works end to end, not claiming a validated fix.

Cost model: no cloud inference, no per-image API fee, no data egress. The marginal cost of processing one more survey is the electricity to run a laptop that is already aboard.

DEPLOYMENT & SCALABILITY — OFFLINE-FIRST BY DESIGN

Laptop

Runs today on an Apple M5, 24 GB — fully offline. No cloud, no per-image cost, no data leaving the vessel.

10.6 MB

ONNX export, measured at 8.5 ms inference — sized for Jetson-class edge hardware aboard an AUV or survey launch.

REST

FastAPI service + SQLite register: the same pipeline scales from one operator console to a shore-side survey fleet.

Not yet

Never run on a Jetson or a live AUV. That is the next milestone, and it is a target — not something we claim today.



Prior art, method, and what we have not done

Judges trust a team that states its limits. The right-hand box is on the slide deliberately.

PRIOR ART — AND OUR RELATIONSHIP TO IT

WORK	LICENCE	RELATIONSHIP
GhostVision JMSE 14(10):951, 2025	NOASSERTION	Closest system. Not vendored — licence unresolved.
PINGMapper Earth & Space Science, 2022	MIT	Pipeline shape and output conventions.
sidescantools	GPL-3.0	Studied; not vendored.
AI4Shipwrecks arXiv 2401.14546	MIT	Route to a wreck class later.
MILCO / NOMBO Data in Brief 53:110132	CC BY 4.0	Training data — mine-like objects.
sss-crab-pot-detection-ds PINGEcosystem	CC BY-SA 4.0	Training data — derelict ghost gear.
SSM-DETR · TR-YOLOv5s MSF-DETR · LEF-RT-DETR 4 detection architectures reviewed	papers	Evaluated, not adopted. Each needs 2.6x–44x our compute for single-digit AP gains, on datasets we cannot access or that are forward-looking sonar, not side-scan.

Why we did not simply adopt a published architecture. The published gains are real — but they are measured on a different sonar modality, at a compute budget that rules out edge deployment, on data we cannot obtain to reproduce. Our bottleneck is false alarms on empty seabed, which a bigger backbone does not fix.

METHOD REFERENCES

Lee (1980) — adaptive speckle filtering, DOI 10.1109/TPAMI.1980.4766994 · **Platt (1999)** — probabilistic outputs for large-margin classifiers, MIT Press · **Guo et al. (2017)** — on calibration of modern neural networks, arXiv:1706.04599 · **Ultralytics YOLO11** — detection backbone.

WHAT WE HAVE NOT DONE — STATED DELIBERATELY

- Ghost-gear detection has **one raw training result** (mAP50 0.323). It has not been through the verification stage, calibration, or an ablation.
- Geolocation accuracy has **never been validated** — the licensed data available to us ships no navigation track.
- Detection **degrades under added speckle noise** (measured). A noise-robust checkpoint exists but trades clean accuracy.
- Never run on a **Jetson or a live AUV**.
- Our own sonar preprocessing chain **did not help** when measured properly, and ships disabled.
- Detector backend is **AGPL-3.0**; the licence is isolated behind an interface but not yet replaced.

ROADMAP TO THE FINALS — HOW EACH GAP CLOSES

1 · Verify	Fit the verification stage on the ghost-gear model and run the full ablation — turn one raw number into a pipeline result.
2 · Compare	Measure against detector-only, rule-filter and YOLO11s / RT-DETR at equal compute.
3 · Calibrate	Report reliability curve and ECE on the held-out split.
4 · Edge	Benchmark the ONNX export on Jetson-class hardware.
5 · Geolocate	Validate positional accuracy against a survey that actually ships a navigation track.

Everything above is reproducible. Every figure in this deck traces to experiments/registry.jsonl, docs/BENCHMARKS.md, or a live run of the prototype — regenerated by script, never typed in by hand.